

RAY OPTICS

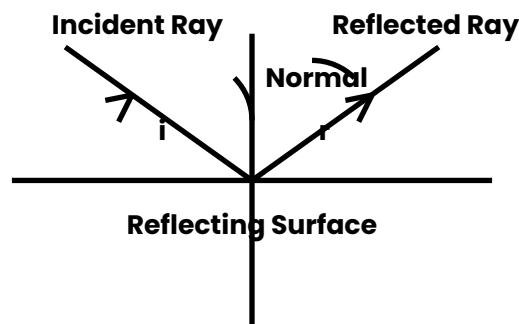
Ray Optics

Reflection of Light

Reflection of light is the phenomenon in which light rays return into the same medium after striking a reflecting surface.

Laws of Reflection

- The incident ray, reflected ray and normal at the point of incidence lie in the same plane.
- The angle of incidence is equal to the angle of reflection.



Law of Reflection

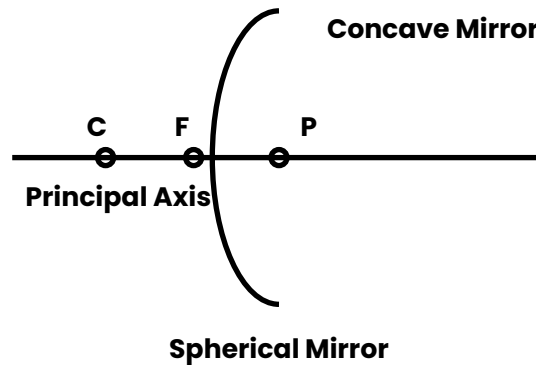
Spherical Mirrors

A spherical mirror is a part of a hollow sphere whose one surface is polished and the other surface is reflecting.

Important Terms

- Pole is the centre of the reflecting surface of a spherical mirror.
- Centre of curvature is the centre of the sphere of which the mirror is a part.
- Radius of curvature is the distance between the pole and centre of curvature.

- Principal axis is the straight line passing through the pole and centre of curvature.
- Principal focus is the point where rays parallel to the principal axis meet or appear to meet after reflection.



Mirror Formula

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

Where:

- f = Focal length
- v = Image distance
- u = Object distance

Relation Between Focal Length and Radius

$$f = \frac{R}{2}$$

Magnification by Mirror

$$m = \frac{h_i}{h_o} = -\frac{v}{u}$$

Where:

- m = Magnification
- h_i = Height of image
- h_o = Height of object

Refraction of Light

Refraction is the change in direction of light when it passes obliquely from one transparent medium to another.

Snell's Law

$$n_1 \sin(i) = n_2 \sin(r)$$

Where:

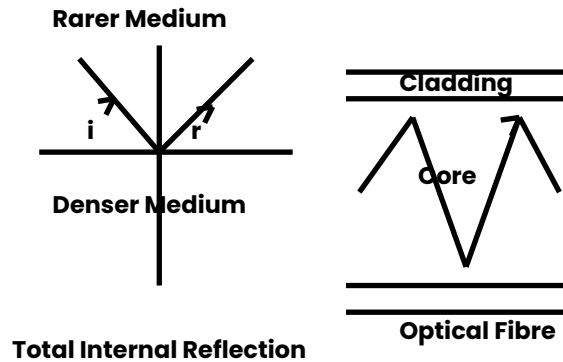
- n_1 = Refractive index of first medium
- n_2 = Refractive index of second medium
- i = Angle of incidence
- r = Angle of refraction

Total Internal Reflection

Total internal reflection occurs when light travelling from a denser medium to a rarer medium is completely reflected back into the denser medium.

Conditions

- Light must travel from an optically denser medium to an optically rarer medium.
- The angle of incidence must be greater than the critical angle.



Critical Angle

$$\sin(C) = n_2 / n_1$$

Where:

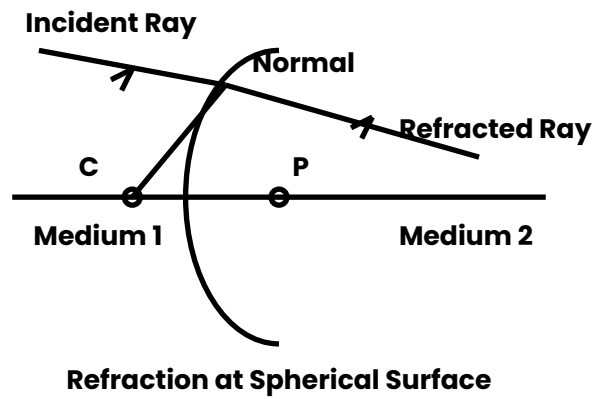
- C = Critical angle
- n_1 = Refractive index of denser medium
- n_2 = Refractive index of rarer medium

Refraction at a Spherical Surface

$$n_2/v - n_1/u = (n_2 - n_1)/R$$

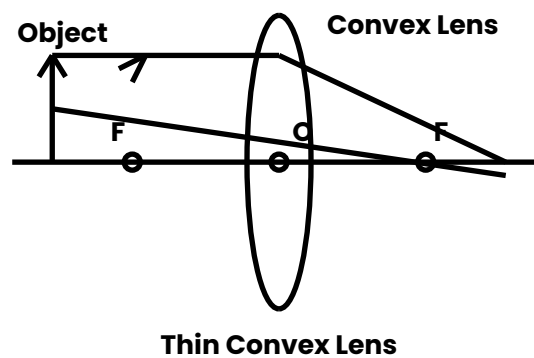
Where:

- n_1 = Refractive index of first medium
- n_2 = Refractive index of second medium
- u = Object distance
- v = Image distance
- R = Radius of curvature



Thin Lenses

A lens is a transparent refracting medium bounded by two surfaces, at least one of which is spherical.



Lens Formula

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

Magnification by Lens

$$m = h_i / h_o = v / u$$

Lens Maker's Formula

$$\frac{1}{f} = (n - 1)\left(\frac{1}{R_1} - \frac{1}{R_2}\right)$$

Where:

- n = Refractive index of lens material
- R_1 = Radius of curvature of first surface
- R_2 = Radius of curvature of second surface

Power of Lens

$$P = \frac{1}{f}$$

Where:

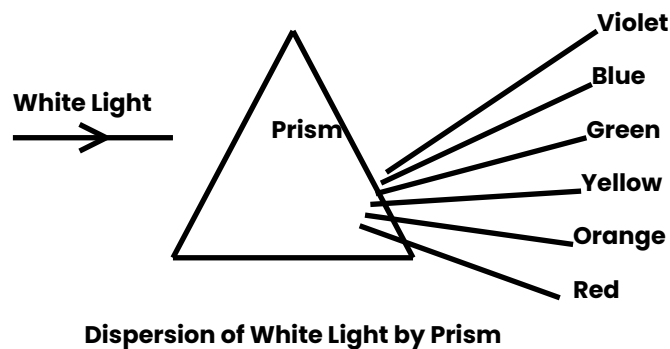
- P = Power of lens in dioptre
- f = Focal length in metre

Combination of Thin Lenses

$$P = P_1 + P_2 + P_3$$

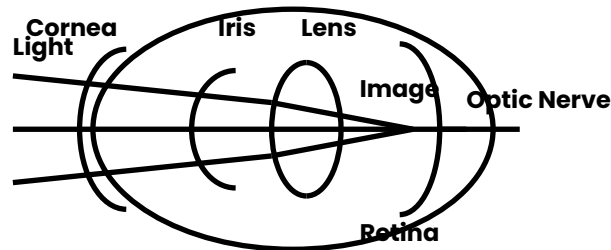
Prism and Dispersion

Dispersion is the phenomenon of splitting white light into its constituent colours when it passes through a prism.



Human Eye

The human eye is a natural optical instrument that forms an image of an object on the retina.



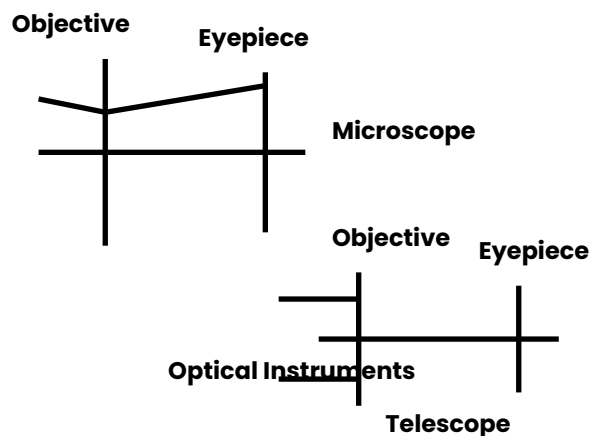
Human Eye

Myopia: image forms in front of retina Hypermetropia: image forms behind retina

Eye Defects

- Myopia is a defect of vision in which distant objects are not seen clearly.
- Hypermetropia is a defect of vision in which nearby objects are not seen clearly.

Optical Instruments



Simple Microscope

A simple microscope is a single convex lens used to obtain a

magnified virtual image of a small object.

Compound Microscope

A compound microscope uses an objective lens and an eyepiece to produce a highly magnified image of a small object.

Telescope

A telescope is an optical instrument used to observe distant objects by producing an enlarged angular view.

Reflecting Telescope

A reflecting telescope uses a concave mirror as the objective to collect and focus light from distant objects.

Quick Revision Points

- Angle of incidence is equal to angle of reflection.
- For a spherical mirror, focal length is half the radius of curvature.
- Refraction occurs when light passes between different transparent media.
- Total internal reflection requires a denser-to-rarer transition and incidence angle greater than the critical angle.
- Lens power is measured in dioptries.
- The power of thin lenses in contact is the algebraic sum of their individual powers.

Important Formulae

- $\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$
- $f = R / 2$

- $m = -v / u$
- $n_1 \sin(i) = n_2 \sin(r)$
- $P = 1/f$
- $P = P_1 + P_2 + P_3$

Conclusion

Ray optics explains the reflection and refraction of light and provides the basic principles used in mirrors, lenses and optical instruments.